

- 4 (a) Fig. 4.1 is a drawing made by a student of a plant cell, showing one of the stages of mitosis.

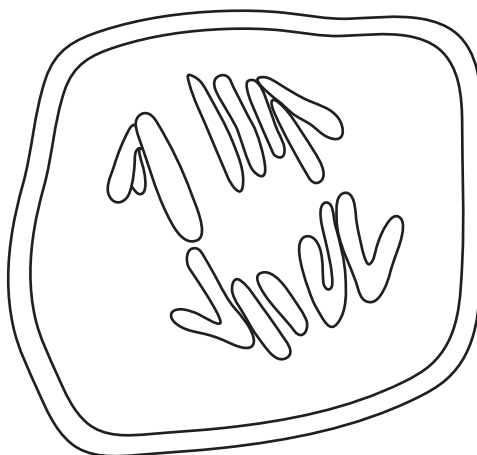


Fig. 4.1

- (i) Describe the role of the spindle during this stage of mitosis.

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- (ii) The chromosomes are composed of DNA.

Describe the structure of a monomer of a DNA molecule.

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- (b) Telomeres are located at the ends of each chromosome.

Telomeres progressively shorten each time a healthy human cell divides, until the cell is no longer able to divide.

Stem cells have an enzyme called telomerase that prevents the telomeres shortening during the cell cycle.

- (i) Suggest why telomerase activity is important in stem cells.

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- (ii) Suggest why tumour cells produce high concentrations of telomerase.

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- (iii) Scientists have researched the possibility of treating cancer by inhibiting the action of telomerase.

One method that has been studied involves inserting small, single-stranded DNA into cancer cells. These sections of DNA have a base sequence that is complementary to a section of the mRNA transcribed from the gene coding for telomerase.

Explain how inserting small, single-stranded DNA could prevent **translation** of the mRNA to produce telomerase.

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[Total: 10]